29. Autocatalytic sets: maximality, irreducibility and dynamical viability

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Could a collection of reactions collectively help produce the substances that make those same reactions happen? Autocatalytic-set theory turns part of that question into a precise network test. Starting from an externally supplied food set, we ask which reactions can be supported by products of the network and which smaller subsets already have that property. This describes a structural prerequisite for self-sustaining chemistry, not a complete definition of life.

Key terms

- **Catalyst** — A substance participating in a reaction mechanism that increases its rate while being regenerated overall; it does not change the equilibrium free-energy difference.
- **Autocatalysis** — Production that directly or collectively helps enable more production of the relevant system.
- **Food set** — The species assumed to be available from outside the network.
- **Closure** — All species producible from the food set using the selected reaction rules; standard RAF closure temporarily ignores catalysis.
- **RAF** — A reflexively autocatalytic and food-generated reaction set: its reactants are reachable from food and each reaction has a catalyst available within the closure.
- **Maximal RAF** — The largest RAF contained in the supplied network, when a RAF exists.
- **Irreducible RAF** — A RAF containing no smaller nonempty RAF subset.
- **Minimum-cardinality RAF** — A RAF with the fewest reactions among all RAFs in the specified network; this is stronger than irreducibility.
- **Chemostat** — An idealized externally maintained concentration or chemical reservoir.
- **Dynamical viability** — The additional question of whether specified reaction rates and conditions support sustained activity or growth.

Why this matters

Collectively supported chemistry is relevant to thinking about how organized reaction systems can arise and persist. Exact minimality results help distinguish essential reactions from dispensable ones in a model. Just as importantly, separating structural closure from energy supply and kinetics prevents a network diagram from being mistaken for proof of a viable replicating system.

What progress would mean

Success would identify supported reaction sets and prove the stated kind of minimality. Demonstrating sustained growth, heredity or evolution would require extra chemical and dynamical assumptions beyond the RAF certificate.

Core question

For a finite catalytic reaction system, can its maximal RAF and smallest RAF subsets be certified, while keeping structural autocatalysis distinct from sustained growth?

Scope and assumptions

Input species S, reactions R with integer stoichiometry, food set F, and catalysis relation $C \subset S \times R$. Begin with at most 12 reactions for exhaustive cross-checks. Catalysis is supplied model data, not inferred from a stoichiometric cycle.

Define three separate targets: unique maximal RAF, an inclusion-minimal (irreducible) RAF, and a minimum-cardinality RAF. Thermodynamic and kinetic viability require extra inputs and are separate extensions.

Mathematical target

For $R' \subset R$ compute cl_{R′}(F) by repeatedly adding products whose reactants are already present, ignoring catalysis in this closure step. R' is a nonempty RAF when every reactant lies in this closure and every reaction has at least one catalyst in it.

An irreducible RAF has no proper nonempty RAF subset. A minimum-cardinality RAF minimizes $|R'|$ over all RAF subsets. Ordinary RAF closure permits catalysts to arise later; a catalysis-respecting startup ordering is a stronger condition and must be tested separately.

Required outputs and proof obligations

Export closure rounds, catalyst witnesses, maxRAF elimination trace, and minimality/cardinality proof with explicit scope. For thermodynamic extension specify chemical potentials, chemostats, and driving. For kinetic extension provide rates, dilution, initial conditions and a positive sustained-state/growth criterion; RAF existence alone is insufficient.

Validation and rejection controls

Cross-check all subsets for tiny networks. Include food-catalyzed, self-catalyzed, catalyst-free, and mutually dependent examples. Verify a maximum RAF can contain several irreducible RAFs of different sizes. Deletion tests for irreducibility must use a complete RAF detector on each remaining network, not merely test whether the entire remainder is itself RAF.

Milestones and research extension

Baseline: checked maxRAF/RAF witnesses. Strong: complete minimum-cardinality results for a bounded network family. Research extension: characterize which RAFs additionally support a declared driven mass-action regime. Biological self-replication and evolution require their own operational definitions.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Hordijk and Steel, Autocatalytic sets in a partitioned biochemical network — RAF closure and the distinction between maximal and irreducible RAFs.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$R' \subseteq R, \quad R' \neq \emptyset, \quad \text{reactants}(r) \subseteq \text{cl}_{R'}(F),$$

$$\forall r \in R' \; \exists s \in \text{cl}_{R'}(F) : \; (s, r) \in C.$$

Compute closure by reactant availability without imposing catalysis in the closure step. These conditions define a RAF. Inclusion-maximal, inclusion-minimal (irreducible), and minimum-cardinality RAFs are separate targets. Catalysis-respecting startup and sustained growth require additional conditions.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — reversed terminology (81).** maxRAF is maximal, not minimal. Inclusion-minimal and minimum-cardinality are also different optimization targets, so the existing output flag is ambiguous.
2. **Major — closure/startup (30–31, 73–79).** Standard RAF food closure ignores catalysis during reachability; requiring every intermediate step to be catalyzed is stronger. The rewrite keeps these questions separate.

3. **Major — thermodynamic viability (99–108).** A reaction list plus arbitrary negative ΔG values is not a consistent driven chemical model. Chemostats, activities and cycle constraints matter; catalysts do not alter equilibrium free-energy differences.
4. **Major — biology and information (97, 112–116).** Generic hypercycle stability depends on the actual equations, and concentrations are not automatically probabilities or a joint distribution for mutual information. Structural RAF certificates cannot alone establish Darwinian evolution, stability against parasites, or information emergence.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a finite catalytic reaction system, can its maximal RAF and smallest RAF subsets be certified, while keeping structural autocatalysis distinct from sustained growth?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: checked maxRAF/RAF witnesses. Strong: complete minimum-cardinality results for a bounded network family. Research extension: characterize which RAFs additionally support a declared driven mass-action regime. Biological self-replication and evolution require their own operational definitions.

Sources supporting the corrections

- Hordijk and Steel, Autocatalytic sets in a partitioned biochemical network — RAF closure and the distinction between maximal and irreducible RAFs.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented

trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.

- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.